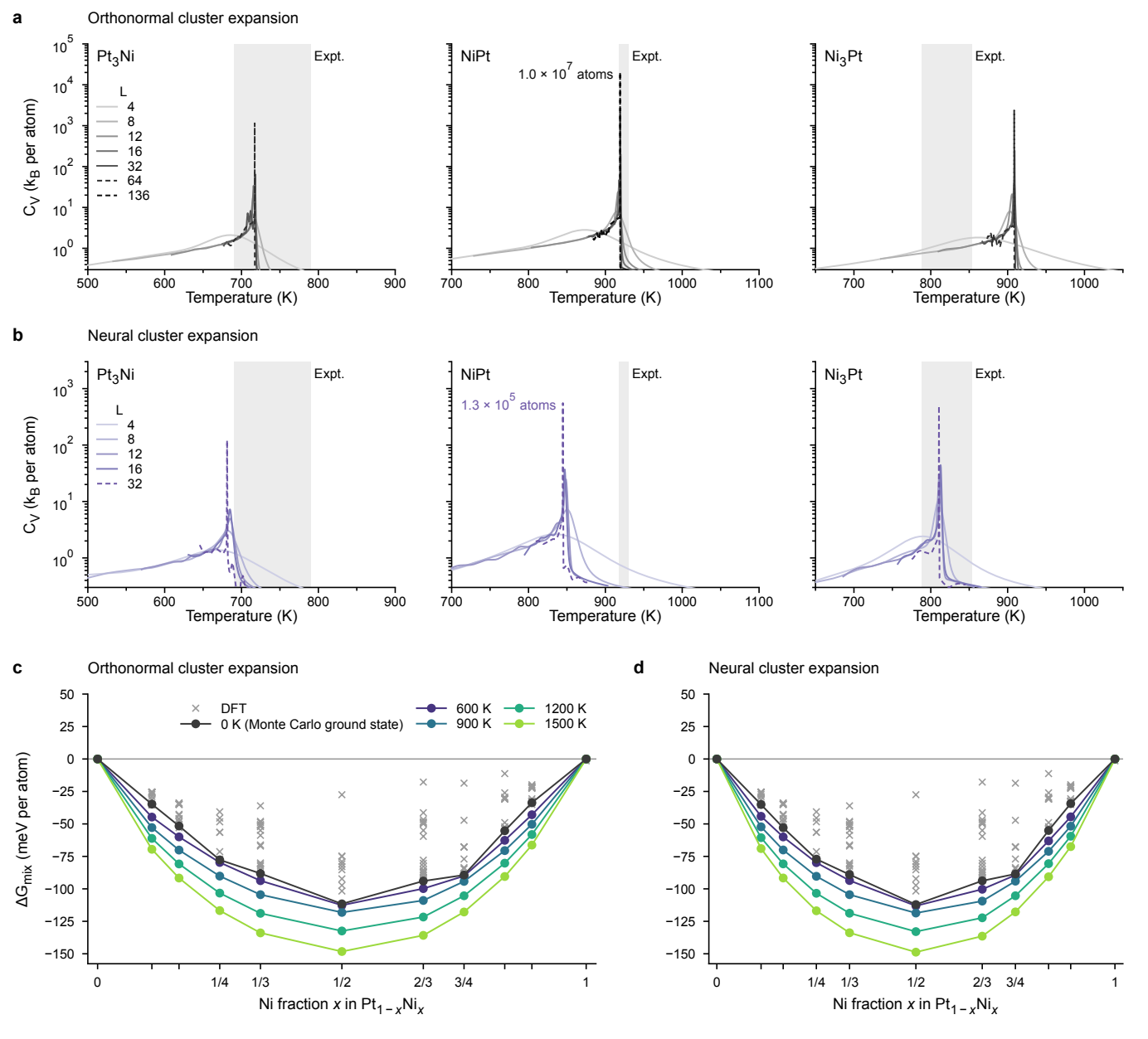

a
Orthonormal cluster expansion
Pt3Ni
Expt.
L
4
8
12
16
32
64
136
NiPt
1.0 × 10^7 atoms
Ni3Pt
CV (kB per atom)
Temperature (K)
b
Neural cluster expansion
1.3 × 10^5 atoms
c
Orthonormal cluster expansion
DFT
0 K (Monte Carlo ground state)
600 K
900 K
1200 K
1500 K
ΔGmix (meV per atom)
Ni fraction x in Pt1−xNix
d
Neural cluster expansion